

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheets for Mainline Fittings
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-004
Document Revision	R01
Document Status	Issued For Review
Originator	Israel Osobase
Security Classification	Restricted
Issue Date	4 Aug 2022

Data Sheets for Mainline Fittings

R01	04.08.22	Issued For Review	I. Osobase	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	04.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

Table of Contents

1. FLANGES 24 INCHES	5
2. FLANGES 36 INCHES	6

1	Flange Nominal Diameter, inch			24 inch (610 mm)
2	Pipeline Service			Corrosive (CO ₂)
3	Type of Effluent			Natural Gas + Condensate
4	Flange Steel Grade			ASTM A694 F70
5	Pressure Rating			900#
6	Flange Nominal Internal Diameter, mm			578.2
7	Design Factor			0.6
8	Standard of Design	Standard of Dimension		MSS SP 44
9		Flange Facing		Ring Type Joint (RTJ)
10	Matching Pipe	Steel Grade		API 5L X70
11		Nominal Internal Diameter, mm		591.2
12		Nominal Wall Thickness, mm		15.88
13		Mill Hydrostatic Pressure		To be Certified to Test Pressure of 90% of SMYS
15	Temperature	Design	Maximum	60° C
16			Minimum	, -13.3° C
19	Design Pressure, barg			110
20	Corrosion Allowance, mm			3mm
21	Charpy V-Notch	Min. Average Value, Joule		56
22		Min. Individual Value, Joule		42
23	Impact Test	Min. Shear Area, %		85
24	Note: Refer to EA-BFPCL-GPMC-GGPL-FEED-PPL-LST-003 - Pipeline MTO for List of Flanges.			

1	Flange Nominal Diameter, inch		36 inch (916 mm)	
2	Pipeline Service		Corrosive (CO ₂)	
3	Type of Effluent		Natural Gas + Condensate	
4	Flange Steel Grade		ASTM A694 F70	
5	Pressure Rating		900#	
6	Flange Nominal Internal Diameter, mm		578.2	
7	Design Factor		0.6	
8	Standard of Design	Standard of Dimension	MSS SP 44	
9		Flange Facing	Ring Type Joint (RTJ)	
10	Matching Pipe	Steel Grade	API 5L X70	
11		Nominal Internal Diameter, mm	591.2	
12		Nominal Wall Thickness, mm	23.83	
13		Mill Hydrostatic Pressure	To be Certified to Test Pressure of 90% of SMYS	
15	Temperature	Design	Maximum	60° C
16			Minimum	, -13.3° C
19	Design Pressure, barg			110
20	Corrosion Allowance, mm			6mm
21	Charpy V-Notch	Min. Average Value, Joule		56
22		Min. Individual Value, Joule		42
23	Impact Test	Min. Shear Area, %		85
24	Note: Refer to EA-BFPCL-GPMC-GGPL-FEED-PPL-LST-003 - Pipeline MTO for List of Flanges.			